

Fundamental Momentum Based Stock Investing

Sicheng Wang Taoying Zhao Pengcheng Zhou

Advisor: Rob Reider, Method Investment & Advisory Ltd

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Background

Extensive Research on Momentum Strategies

- Explaining the profitability of momentum strategies ([Chan et al., 1996](#))
- Exploring the effectiveness of momentum strategies across diverse asset classes ([Asness et al., 2013](#))
- Constructing momentum factor to describe the cross-sectional variation of commodity returns ([Bakshi et al., 2019](#))

Motivation

Recent Development in Momentum Strategies

- The traditional momentum is based on prices (i.e. **price momentum**).
- **Basis-momentum** factor proposed by [Boons and Prado \(2019\)](#) shows effective performance on predicting commodity premiums in both the time series and cross-section.
- **Macro momentum** strategies developed by [Brooks \(2017\)](#) demonstrates consistent performance over a half-century.
- **Fundamental momentum** strategy proposed by [Novy-Marx \(2015\)](#) provides more stable and significant performance compared with price momentum.

Main Results

Goal of Our Project

[Novy-Marx \(2015\)](#) looks at one form of fundamental momentum, earnings per share momentum, whereas we mainly focus on another form of fundamental momentum, profit margin momentum, so that we can provide a better understanding of the momentum profile.

Empirical Interests

- Strategy performance in back-testing framework.
- Spanning tests.
- Relation with traditional price momentum strategies.
- Conditional strategies.

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Data

Data Sources: WRDS

- **Monthly Stock Data:** CRSP database.
- **Quarterly Company Data:** Compustat database.

Data Process

- **Time Range:** from January 1975 to December 2023.
- Use CUSIP to match the stock price with the fundamental data (24260 stocks in the panel).
- Negative prices are dropped.

Methodology: Backtesting Framework

Quarterly-rebalanced Sorted Portfolio

- At each March, June, September, and December, we construct 10 sorted (equally-weighted) portfolios based on the selected fundamental momentum and hold for 3 months.
- If the momentum data is missing when we rebalance the portfolio, we use the most recent data up to two months.
- Considering the liquidity, we exclude the stocks with the lowest 30% volume.
- Transaction costs: 20 basis points for long-only portfolio and 40 basis points for long-short portfolios when rebalancing. (a rough but relatively conservative estimate based on [Frazzini, Israel, and Moskowitz \(2015\)](#))

Methodology: Profit Margin Momentum

Why Profit Margin?

- Profit margin (PM) is a financial metric that indicates how much of a company's revenue remains as profit after covering all expenses.

$$PM = \frac{\text{Net Income}}{\text{Revenue}} \quad (1)$$

- A higher profit margin indicates a more profitable company that has better control over its costs compared to its competitors.

Methodology: Profit Margin Momentum

Factor Construction

$$\text{PM Momentum}(t) = \frac{PM(t) - PM(t - 3)}{SD(PM_{\text{Last } 3\text{Yrs}})} \quad (2)$$

- To increase the stability, we standardize the profit margin momentum by the standard deviation of the profit margin over the last 3 years.
- Moreover, we require that there are at least 6 observations in the last 3 years to calculate the standard deviation.

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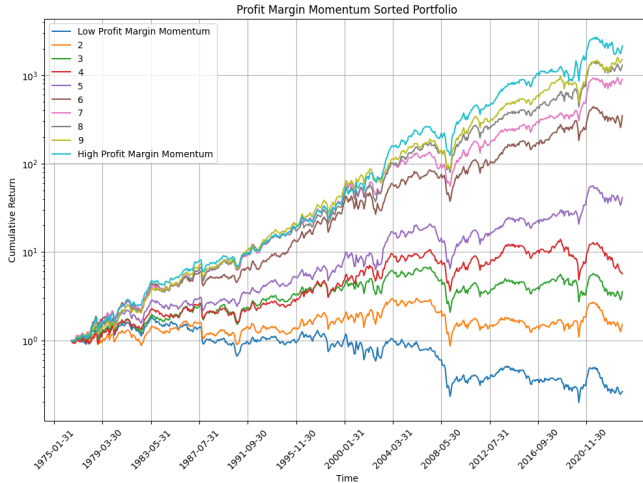
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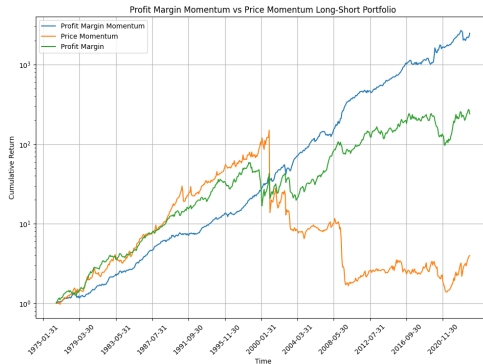
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Sorted Portfolio based on Profit Margin Momentum



Long-short Portfolio based on Profit Margin Momentum



	PM Momentum	Price Momentum	PM
Mean	17.34%	9.58%	14.06%
Std	12.55%	29.27%	22.48%
SR	1.38	0.33	0.63
MDD	25.70%	99.07%	71.53%

Spanning Test

Regression Settings

- Regress profit margin momentum on FF3 factors and price momentum.
 - ▶ Examine whether the profit margin momentum can be subsumed.

Regression Results: $R^2=2.40\%$

	α	$MKT - R_f$	SMB	HML	MOM
Coefficient	0.010	-0.0181	0.0413	0.1761	0.0173
t-statistic	(6.749)	(-0.508)	(0.772)	(3.441)	(0.950)

Strategy Overlap: Definition

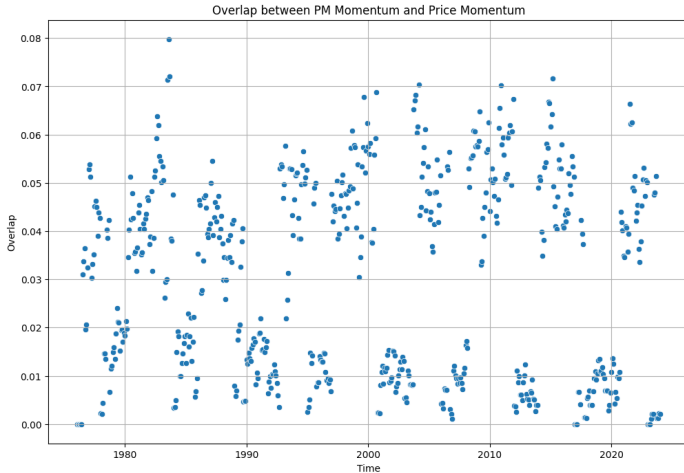
Overlap Ratio

- At time t , for the i -th portfolio of profit margin momentum and price momentum, we calculate the overlap ratio simply by

$$\text{Overlap Ratio} = \frac{\# \text{ of Overlap Stocks}}{\# \text{ of Stocks (Union)}}$$

- For the long-short portfolios, we just take the average of the overlap ratio over the highest and lowest decile portfolios.

Strategy Overlap: Results



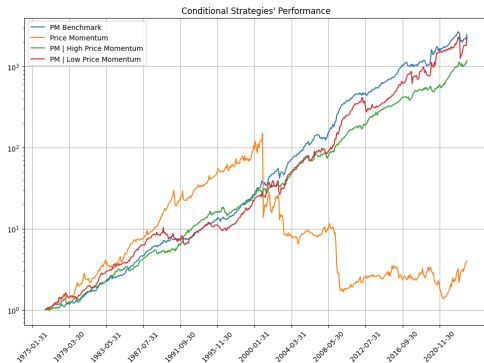
Conditional Strategy

What can be improved?

We want to control the effect of price momentum while building long-short profit margin momentum portfolio. We build long-short double sorted portfolios conditioning on PM momentum and price momentum. We compared the following two long-short portfolios.

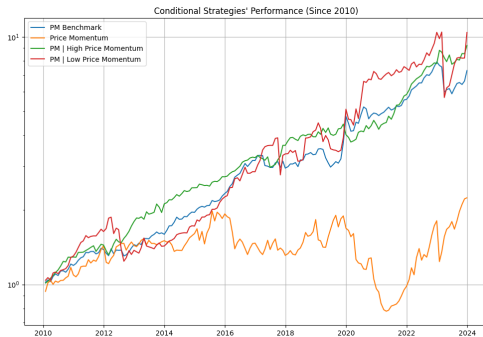
- High PM | High Price – Low PM | High Price
- High PM | Low Price – Low PM | Low Price

Conditional Strategy: Results



	PM Momentum	High Price M.	Low Price M.
Mean	17.34%	15.68%	18.43%
Std	12.55%	11.72%	19.60%
SR	1.38	1.34	0.94
MDD	25.70%	21.79%	45.56%

Conditional Strategy: Results(Since 2010)



	PM Benchmark	High Price PM.	Low Price PM.
Mean	15.39%	16.65%	20.59%
Std	15.03%	11.73%	26.25%
SR	1.02	1.42	0.78
MDD	25.70%	15.36%	45.56%

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Conclusion

- Profit margin momentum strategy provides a more stable and significant performance compared with price momentum.
- The profit margin momentum strategy is not subsumed by the traditional Fama-French 3 factors and price momentum.
- Though the overlap of the profit margin momentum and price momentum strategies is relatively low, the conditional strategy generates a more stable performance by reducing the effect of price momentum.

Thank You!